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A spectral index for the detection of algal blooms using Sentinel-2 Multispectral Instrument (MSI) imagery: a case study of Hulun Lake, China

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ABSTRACT

Lakes at a global level have increasingly experienced algal blooms in recent decades, and it has become a key challenge facing the aquatic ecological environment. Remote sensing technology is considered an effective means of algal bloom detection. This study proposed a novel algal bloom detection index (ABDI) based on Sentinel-2 Multispectral Instrument (MSI) data. The ABDI was evaluated by application to Hulun Lake, China. Areas of algal bloom detected by the ABDI were consistent with those identified from visual interpretation maps [the coefficient of determination = 0.87; root-mean-square error = 0.67 km²; overall accuracy >98%; Kappa coefficient >0.88]. The ABDI was less sensitive to thin cloud and turbid water compared to the floating algae index (FAI), adjusted floating algae index (AFAI), normalized difference vegetation index (NDVI), and enhanced vegetation index (EVI). Algal bloom dynamics in relation to meteorological factors in Hulun Lake were analysed using time-series MSI data, which indicated that algal blooms occurred mainly in summer and were distributed in the near-shore waters. Temperature, precipitation, sunshine duration, and wind speed as well as human activities were found to influence spatio-temporal patterns of algal blooms. The results indicate that ABDI is applicable to the detection of algal blooms under a variety of environmental conditions occurring in other regions, such as in the Taihu, Dianchi, and Chaohu lakes and the Yellow Sea. The results of this study can provide an operational algorithm for the detection of algal blooms and environmental management.

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1. Introduction

Lakes constitute an important component of global water resources, and play significant roles in water cycling, climate regulation, and biodiversity protection. However, lakes are sensitive to environmental changes, and lakes globally are increasingly being negatively impacted by algal blooms due to climate change, rapidly increasing populations and unsustainable utilization of water resources (Ho, Michalak, and Pahlevan 2019). Incidents of algal blooms are indications of lake eutrophication, and the inputs of excess nutrients

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into water resources pose a serious threat to the aquatic ecological environment (Paerl, Hall, and Calandrino 2011). More specifically, large-scale algal blooms may lead to the destruction of the ecological landscape, biological hypoxia, and the accumulation of toxins, which may further affect tourism and fisheries (Fang et al. 2018a). Toxic algal blooms pose a threat to human health through the food chain. Therefore, the monitoring of algal blooms is of great significance for the protection and management of water resources and aquatic ecosystems as well as for protecting human health.

Remote sensing technology has become an effective method for the monitoring of algal blooms. Within the use of this technology, spectral indices are most commonly used for extracting information for algal blooms from images due to their easy implementation and flexibility. Spectral indices are generally expressed as the combination of visible or infrared bands. There have been to date many studies on the development of spectral indices for the monitoring of algal blooms (Sagan et al. 2020; Shi et al. 2019; Xiong et al. 2019). Rouse et al. (1973) proposed the concept of the normalized difference vegetation index (NDVI), and Huete and Justice (1999) used the NDVI to develop an enhanced vegetation index (EVI). However, previous studies have found that the NDVI and EVI are easily disrupted by the atmosphere, cloud cover, and turbid surface waters (Chen et al. 2014; Li, Niu, and Jiang 2010). Hu (2009) developed the floating algae index (FAI), which has been successfully applied to many lakes and marine regions for the monitoring of macro or micro-algal blooms (Chen et al. 2019; Huang et al. 2013; Hu et al. 2010; Naghdi et al. 2018; Oyama, Matsushita, and Fukushima 2015; Zhang et al. 2015). Afterwards, Fang et al. (2018a) proposed an adjusted floating algae index (AFAI). The FAI and AFAI have been shown to be less sensitive to aerosols, solar/viewing geometry, and sun glint compared with the NDVI and EVI (Zhang et al. 2019). However, the use of the FAI can lead to the incorrect interpretation of turbid waters as algal blooms (Liu et al. 2015). In addition, some studies have shown that the combination of different spectral indices can successfully be applied to distinguish between algal blooms and aquatic vegetations in complex water ecosystems (Liang et al. 2017; Qing et al. 2020; Shen et al. 2019; Zhou et al. 2018). Unfortunately, these indices were designed to distinguish between algal blooms occurring in specific areas, and many uncertainties remain in their application to other regions.

Spectral indices, including those mentioned above, can be readily applied to the data from many different remote sensing technologies, including Land Remote-Sensing Satellite (Landsat), Huanjing-1A/B (HJ-1A/B), Moderate Resolution Imaging Spectroradiometer (MODIS), Sea-viewing Wide Field Sensor (SeaWiFS) and the Medium Resolution Imaging Spectrometer (MERIS) (Ackleson and Klemas 1987; Binding, Greenberg, and Bukata 2013; Gower et al. 2006; Mishra and Mishra 2012; Shanmugam, Ahn, and Ram 2008; Xing and Hu 2016). However, limitations remain in the use of these data for the monitoring of algal blooms in complex lake waters. The use of Landsat data for high-frequency monitoring remains impossible due to an insufficient temporal resolution (16 days). Although MODIS, SeaWiFS, and MERIS have high temporal and spectral resolutions, data from these technologies cannot be used to derive detailed information for features with small spatial extents, such as an estuary or lake, due to the coarse spatial resolution of these data. Some indices such as FAI are not suitable for HJ-1A/B data due to limited spectral bands. In contrast, the Sentinel-2 Multispectral Instrument (MSI) is a suitable sensor for the detection of algal blooms occurring over a small area, with a 10 to 60 m spatial resolution, 13 bands

in the range of 400 to 2,500 nm, and a higher temporal resolution of 2 to 5 days. More specifically, the MSI incorporates three red edge bands (705 nm, 740 nm, and 783 nm) in the spectral range of 700 to 800 nm. Although the MSI technology has opened the potential for remote monitoring of algal blooms, relatively few studies to date, have used this technology for this purpose (Soria-Perpinya et al. 2020). Therefore, there is scope for the establishment of a novel algal bloom index for use with the MSI sensor data and for the evaluation of this index by application to case studies.

The aims of the present study were to develop a spectral index for the monitoring of algal blooms and to evaluate this index by application to a case study lake. The objectives of the present study were to: (1) develop an algal bloom detection index (ABDI) based on the spectral features of algal blooms and non-algal blooms in Sentinel-2 MSI images; (2) compare the ability of ABDI to detect algal blooms in Hulun Lake with those of the FAI, AFAl, NDVI and EVI indices; (3) evaluate the importance of spatial resolution on the accuracy of extracted algal bloom information. The effects of atmospheric correction, and the spatial and temporal distributions of algal blooms in Hulun Lake and their relationships to meteorological factors were also discussed in the present study.

2. Materials and methods

2.1. Study area: Hulun Lake

Hulun Lake is not only an important ecological buffer zone in northern China but also plays an important role in sustainable economic development (Fang et al. 2018b). Unfortunately, the water quality of Hulun Lake has gradually deteriorated due to a warming climate and human activities, and lake eutrophication has resulted in frequent algal blooms in recent decades (Chen et al. 2012; Chuai et al. 2012; Guo et al. 2019a, 2019b; Wang, Li, and Sun 2014; Wu et al. 2012). Thus, the monitoring of algal blooms in Hulun Lake is of significance for improving management of the lake.

Hulun Lake (48°33' to 49°20' N, 116°58' to 117°48' E) is located in the cold and semi-arid regions of the Inner Mongolia Autonomous Region, China (Figure 1), and is the largest lake in the Inner Mongolia Autonomous Region, with an area of about 2,063 km² in 2019 and an average water depth of 5 to 6 m (Chuai et al. 2012). The study area falls within a temperate continental monsoon climate zone characterized by warm and semiarid summers and cold and long winters. Hulun Lake remains frozen for 6 to 7 months every year from November to early May of the following year. The sources of water feeding Hulun Lake mainly include precipitation, groundwater and inflows from the Kherlen and Urson rivers (Wang et al. 2015). The main phytoplankton occurring in Hulun Lake includes Cyanophyta, Chlorophyta, and Bacillariophyta. Cyanophyta dominate over the growing season, with major genera including *Microcystis* Kütz., *Anabaena* Bory., and *Aphanizomenon* Morr. The dominant genera of Chlorophyta and Bacillariophyta occurring in the lake include *Oocystis* Näg, *Crucigenia* Morr, *Navicula* Bory, and *Cyclotella* Kütz (Jiang et al. 2011).

2.2. Satellite data and pre-processing

The present study used Sentinel-2 Level-1 C Top-Of-Atmosphere (TOA) satellite images for the detection of algal blooms in Hulun Lake. The Sentinel-2 A and B satellites were

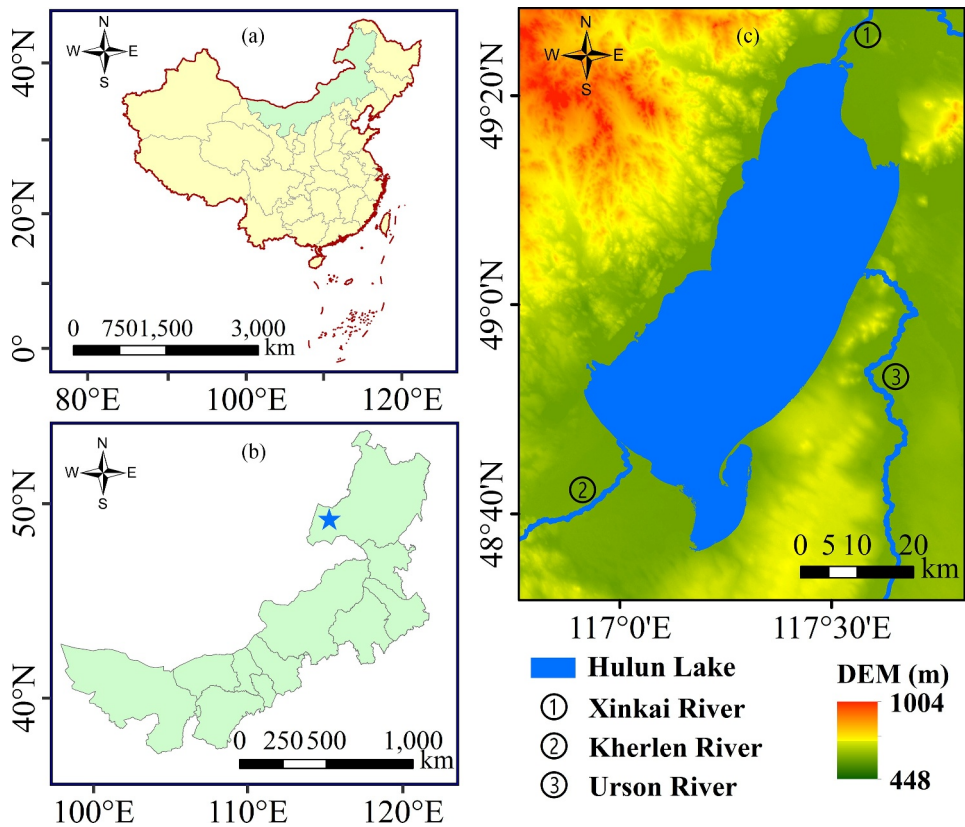


Figure 1. Map showing the location of Hulun Lake (c), within the Inner Mongolia Autonomous Region (b), China (a).

launched by the European Space Agency (ESA) on 23 June 2015 and 7 March 2017, respectively. Sentinel-2 is equipped with a high-resolution multispectral sensor with 13 spectral bands, including visible, red edge (RE), near-infrared (NIR) and shortwave infrared (SWIR) bands (Table 1) (Main-Knorn et al. 2015). The Sentinel-2 MSI has a finer spatial resolution of 10 to 60 m, shorter revisit period, or temporal resolution of 5 days near the equator and 2 to 3 days at mid-latitude areas and a higher spectral resolution of 13 bands compared to other satellite imagery technologies.

The present study processed 13 MSI images of algal blooms from Hulun Lake, including five, five and three images occurring in July, August and September 2019, respectively. Atmospheric correction was applied using the Sen2Cor algorithm (ESA 2015), which is a Level-2A processor that corrects the effects of the atmosphere within Sentinel-2 Level-1 C TOA images to produce the Level-2A surface reflectance product (Mayer and Kylling 2005). The output of Sen2Cor has spatial resolutions of 60 m, 20 m, and 10 m. Surface reflectance (R) of each band was multiplied by 0.0001 and resampled to 10 m using the ENVI 5.3 software (Page, Olmanson, and Mishra 2019). Pixels obscured by thick clouds were masked using the $R(560) > 0.2$ method (Fang et al. 2018a). The Modified Normalized Difference Water Index (MNDWI) was used to identify the boundaries of the lake (Xu

Table 1. Sensor information for the Sentinel-2 MSI, including band, wavelength centre, bandwidth and spatial resolution (Li et al. 2017).

Band ID	Band	Wavelength centre (nm)	Bandwidth (nm)	Spatial resolution (m)
B1	Aerosols	443	20	60
B2	Blue	490	65	10
B3	Green	560	35	10
B4	Red	665	30	10
B5	Red-edge 1 (RE ₁)	705	15	20
B6	Red-edge 2 (RE ₂)	740	15	20
B7	Red-edge 3 (RE ₃)	783	20	20
B8	Near Infrared (NIR)	842	115	10
B8A	Near Infrared-narrow (NIR _n)	865	20	20
B9	Water-vapour	945	20	60
B10	Cirrus	1,375	30	60
B11	Shortwave Infrared 1 (SWIR ₁)	1,610	90	20
B12	Shortwave Infrared 2 (SWIR ₂)	2,190	180	20

2005), and if the image was obscured by thick clouds, the boundary of the lake was determined from an image taken recently.

The present study also used the Landsat-8 Operational Land Imager (OLI) surface reflectance product and MODIS surface reflectance product as auxiliary data for the evaluation of atmospheric correction and verification of algal blooms detected through MSI.

The present study further analysed the factors influencing the changes in algal bloom area in 2019 based on the meteorological data, including temperature, precipitation, duration of sunshine and wind speed. Meteorological data were obtained from the China Meteorological Data Network. Due to the lack of measured hydrological data, the impact of hydrological parameters on algal bloom area has not been studied.

2.3. Methods

2.3.1. Analysis of spectral features

Previous studies often misclassified turbid water as incidents of algal blooms (Liu et al. 2015). This was avoided in the present study by differentiating between turbid and clear water types. Algal blooms occurred in Hulun Lake in 24 and 31 August 2019. The present study selected 1,200 random sample pixels from standard false colour images (RGB: B8A/B4/B3) for each water type, namely algal blooms, clear water, and turbid water. More specifically, samples for algal blooms, clear water, and turbid water were selected from typical algal blooms, cloudless clean water (in the middle of the lake), and estuarine areas of the Kherlen River, respectively (Figure 1).

Figure 2 shows the mean and standard deviation of spectral reflectance, where it is evident that the spectral curve of algal blooms was similar to that of terrestrial vegetation. A small reflectance peak at the green band (560 nm) and an absorption valley at the red band (665 nm) were evident due to contents of chlorophyll-a and other pigments (Zhou et al. 2004) and phycocyanin absorption (Li et al. 2009; Zhu et al. 2016). The algal blooms showed high reflectance peaks in the spectral region of 740 to 865 nm and the lowest reflectance at SWIR bands (Duan, Zhang, and Zhang 2008; Li et al. 2018). Some overlaps between the reflectance of algal blooms in the region of RE₁ to SWIR₂ with that of turbid

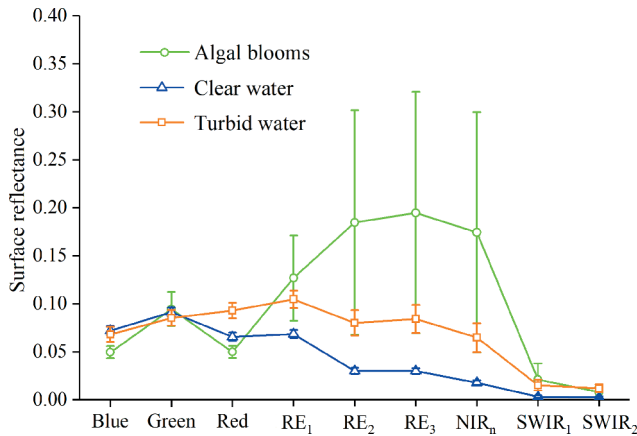


Figure 2. Mean and standard deviation of spectral reflectance for algal blooms (green circles), clear water (blue triangles) and turbid water (yellow squares) from atmospheric corrected Sentinel-2 MSI images for Hulun Lake, China, acquired on 24 and 31 August 2019.

water were evident. The reflectance peak of clear water was located at the green band, and reflectance decreased with increasing wavelength. However, the maximum reflectance for turbid water appeared at the RE₁ band (705 nm). The spectral reflectance of turbid water was higher than that of clear water in the range of red to SWIR bands due to the impacts of suspended particulate matter. More specifically, the red band reflectance of turbid water was higher than those of algal blooms and clear water.

2.3.2. Algal bloom detection index: ABDI

The present study established ABDI for the MSI sensor for identifying algal blooms in Hulun Lake based on the spectral characteristics of different types of water outlined in Section 2.3.1. As shown in Equation (1), the ABDI consists of two parts. The first part is mainly used to extract algal blooms, which is expressed as the difference between RE₂ reflectance and a baseline between red and NIR_n bands (Figure 3 (a) – (d)). Although this aspect of ABDI could distinguish between algal blooms and clear water, some overlaps between turbid water and algal blooms existed (Figure 3 (d)). Accordingly, the second part of ABDI was designed to eliminate the impact of turbid water, which is achieved by subtracting half of the green band reflectance from the red band (Figure 3 (a) – (d)). As shown in Figure 3 (d), the second part of the ABDI value of turbid water exceeded those of algal blooms and clear water. The value of turbid water is then greatly reduced by subtracting the second part from the first part, thereby allowing algal blooms to be distinguished from turbid water (Figure 3 (d)).

$$ABDI = [R_{RE2} - R_{Red} - (R_{NIRn} - R_{Red}) \times (\lambda_{RE2} - \lambda_{Red}) / (\lambda_{NIRn} - \lambda_{Red})] - [R_{Red} - 0.5 \times R_{Green}] \quad (1)$$

In Equation (1), R_{Green} , R_{Red} , R_{RE2} and R_{NIRn} represent the reflectance at green, red, RE₂ and NIR_n bands, respectively, and λ_{Red} , λ_{RE2} and λ_{NIRn} represent the central wavelengths of red, RE₂ and NIR_n bands, respectively.

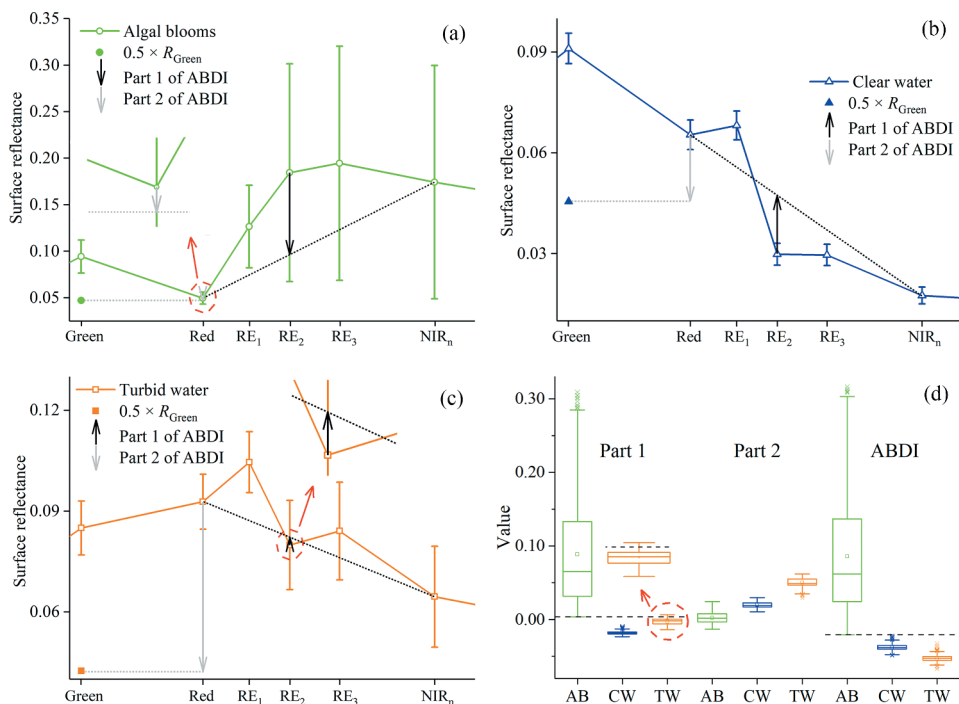


Figure 3. Diagrams describing ABDI: (a) algal blooms (AB); (b) clear water (CW); (c) turbid water (TW); (d) box-and-whisker plots of ABDI and part 1 and part 2 of the index. Note that the red dashed line indicates the location of the enlarged area. The black and grey dot-dash lines in (a), (b) and (c) represent the baselines of part 1 and part 2 of the index, respectively. The black dashed line indicates the minimum value of algal blooms.

2.3.3. Determination of the threshold for the ABDI

The present study used the Otsu algorithm to determine the threshold of ABDI for distinguishing algal blooms (Otsu 2007), and the performance of this approach was good for most of the images. The approach used to apply the Otsu algorithm is detailed below. The image was divided into two classes, and a threshold value corresponding to the maximum variance of the two classes was then used to distinguish between algal blooms and open water. Visual inspection of several MSI images with poor extraction results allowed further adjustment of the threshold value to obtain optimal algal bloom extraction results. In contrast to waters with an absence of algal blooms, waters containing floating algal blooms have a higher reflectance in the NIR band. Therefore, algal blooms could be clearly distinguished from surrounding waters by visual inspection of the standard false colour composite (RGB: B8A/B4/B3) of the MSI image after interactive colour stretching (Hu 2009).

2.3.4. Accuracy assessment

The present study also compared the results of the ABDI with those of FAI, AFAI, NDVI and EVI. These indices were calculated as follows:

$$FAI = R_{NIR} - [R_{Red} + (R_{SWIR} - R_{Red}) \times (\lambda_{NIR} - \lambda_{Red}) / (\lambda_{SWIR} - \lambda_{Red})] \quad (2)$$

$$AFAI = R_{NIR} - R_{Red} + (R_{SWIR} - R_{Red}) \times 0.5 \quad (3)$$

$$NDVI = (R_{NIR} - R_{Red}) / (R_{NIR} + R_{Red}) \quad (4)$$

$$EVI = G \times (R_{NIR} - R_{Red}) / (R_{NIR} + C_1 \times R_{Red} - C_2 \times R_{Blue} + C_3) \quad (5)$$

where R_{Blue} , R_{Red} , R_{NIR} and R_{SWIR} represent the surface reflectance of the blue, red, NIR_n and SWIR₁ bands, respectively, λ_{Red} , λ_{NIR} and λ_{SWIR} represent the central wavelengths of the red, NIR_n and SWIR₁ bands, respectively, G is the gain factor, and C_1 , C_2 and C_3 are empirical coefficients ($G = 2.5$, $C_1 = 6$, $C_2 = 7.5$, $C_3 = 1$) (Hu 2009; Fang et al. 2018a).

The results of the spectral index classification were validated against a visual interpretation map. The confusion matrix between results from visual interpretation and ABDI was established and the overall accuracy, Kappa coefficient (K), commission error, and omission error were calculated. In addition, the coefficient of determination (R^2) and root-mean-square error (RMSE) between visually interpreted and ABDI extracted results were calculated based on randomly selected grid cells ($3 \text{ km} \times 3 \text{ km}$) in the study areas. R^2 and RMSE are expressed as follows:

$$R^2 = \frac{\left[\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}) \right]^2}{\sum_{i=1}^N (x_i - \bar{x})^2 \sum_{i=1}^N (y_i - \bar{y})^2} \quad (6)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - y_i)^2} \quad (7)$$

where N is the number of samples, x is the predicted value, and y is the observed value.

3. Results

3.1. Evaluation of the results of algal bloom detection

Figure 4 and Figure 5 show the standard false colour composite MSI image and the results of algal bloom detection from visual interpretation and spectral indices (ABDI, FAI, AFAI, NDVI and EVI) on 24 and 31 August 2019. Turbid water was noted in the Kherlen River estuary and some thin cloud cover appeared over the southern part of the lake on 24 August 2019 (Figure 4). In comparison to the visual interpretation, it was found that ABDI, FAI, AFAI, NDVI and EVI effectively allowed the identification of algal blooms in images with no cloud cover. The majority of algal blooms appeared as strips and patches, and were mainly distributed in the northern, middle and eastern parts of the lake. The Kherlen River estuarine area (red square in Figure 4) in which algal blooms, thin cloud cover, and turbid water coexisted was enlarged to further explore the ability of each index to distinguish algal blooms under conditions of turbid water and thin cloud cover. The results indicated good agreement between ABDI and visual interpretation, whereas the other indices assessed did not perform as well, with the FAI, AFAI, NDVI and EVI indices misclassifying thin cloud cover and/or turbid water as algal blooms to differing extents.

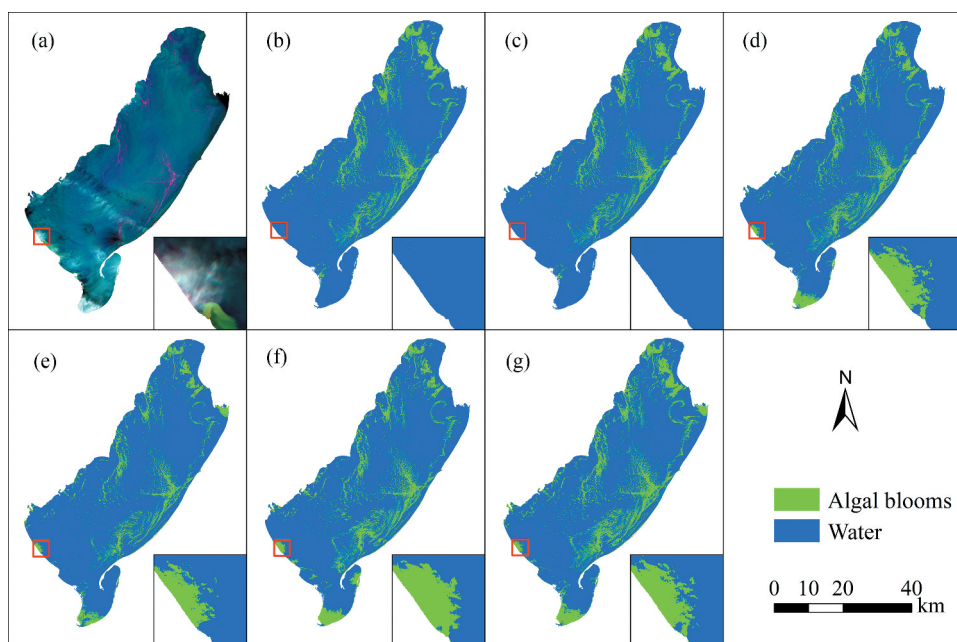


Figure 4. Standard false colour composite (RGB: B8A/B4/B3) of the Sentinel-2 MSI image (a), visual interpretation map (b) and results of algal bloom extraction based on ABDI (c), FAI (d), AFAI (e), NDVI (f) and EVI (g) on 24 August 2019 for Hulun Lake, China. Note that red square is the location of the enlarged map.

On 31 August 2019, algal blooms were mainly distributed in the northern, western and eastern littoral areas of the lake. Turbid water was noted in the Kherlen River estuary and large areas of thin cloud cover were distributed in the southern part of the lake (Figure 5). As shown in Figure 5, the spatial distribution of algal blooms extracted by ABDI was clearly consistent with the results of visual interpretation. In contrast, obvious differences between the areas of algal blooms identified by the other spectral indices and that through visual interpretation were evident, particularly for regions affected by turbid waters and thin cloud cover. NDVI showed the largest number of misclassified pixels, followed by EVI, FAI and AFAI (enlarged part in Figure 5). These results indicated that the ability of the ABDI to correctly identify algal blooms within areas of turbid water or thin cloud cover was higher compared to the abilities of the other assessed indices.

Table 2 shows the overall accuracy, K and commission/omission errors of the spectral indices. The ABDI showed a good performance in detection of algal blooms, achieving the highest overall accuracies of 99.60% and 98.59%, K of 0.95 and 0.89 and the lowest commission errors of 0.00% and 18.21% and omission errors of 8.86% and 0.21% on August 24 and 31, respectively. Figure 6 shows the results of a comparison between algal bloom areas detected through spectral indices and that identified through visual interpretation, where it is evident that the former was in good accordance with the latter, achieving an R^2 and RMSE of 0.87 and 0.67 km², respectively. In contrast, the same comparison for the other indices yielded less favourable results, with the AFAI, FAI, EVI and NDVI achieving an R^2 of 0.63, 0.60, 0.58 and 0.20, respectively, and an RMSE of 1.12 km², 1.52 km², 1.58 km² and 2.26 km²,

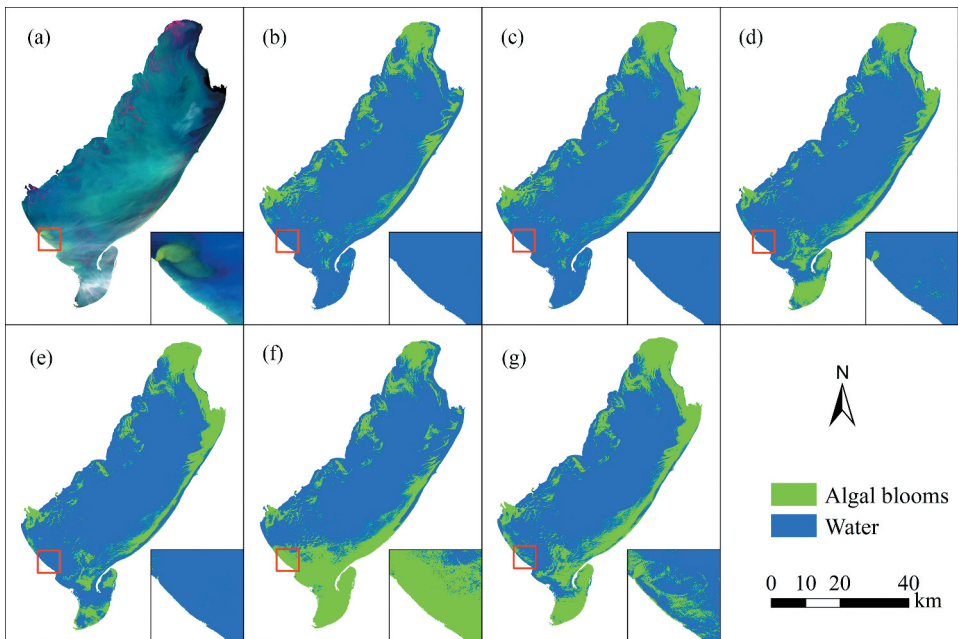


Figure 5. Standard false colour composite (RGB: B8A/B4/B3) of the Sentinel-2 MSI image (a), visual interpretation map (b) and results of algal bloom extraction based on ABDI (c), FAI (d), AFAI (e), NDVI (f) and EVI (g) on 31 August 2019 for Hulun Lake, China. Note that red square is the location of the enlarged map.

Table 2. The performance of ABDI, FAI, AFAI, NDVI and EVI for extraction of algal blooms on 24 and 31 August 2019 for Hulun Lake, China.

Date	Method	Overall accuracy, OA (%)	K	Commission error, CE (%)	Omission error, OE (%)
24 August 2019	ABDI	99.60	0.95	0.00	8.86
	FAI	98.03	0.78	24.28	16.71
	AFAI	97.91	0.74	19.07	29.38
	NDVI	97.34	0.72	32.73	19.62
	EVI	97.49	0.76	34.50	5.68
31 August 2019	ABDI	98.59	0.89	18.21	0.21
	FAI	95.06	0.65	42.34	19.69
	AFAI	95.33	0.65	40.13	22.33
	NDVI	90.63	0.44	62.94	29.50
	EVI	92.99	0.58	53.12	11.32

respectively. These results indicated that the ABDI showed a greater capability in detecting algal blooms in Hulun Lake compared to the other indices assessed.

3.2. Sensitivity analysis

The robustness of the ABDI was further assessed in the present study by investigating the capability of the ABDI to extract algal bloom information under different environmental conditions, such as clear water, turbid water, and thin cloud cover. MSI images from 24 to 31 August 2019 containing algal blooms, clear water, turbid water, and thin cloud cover were randomly sampled, following which, as shown in

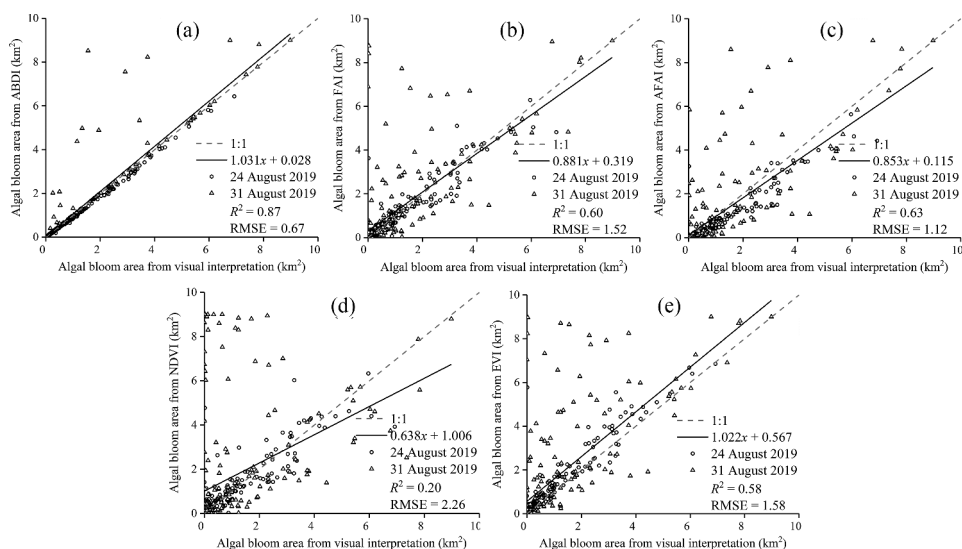


Figure 6. Comparison between areas of algal blooms derived from visual interpretation and that derived through ABDI (a), FAI (b), AFAI (c), NDVI (d) and EVI (e) on 24 and 31 August 2019 for Hulun Lake, China.

Figure 7, the ABDI, FAI, AFAI, NDVI and EVI values for these images were calculated and compared. The results showed that the minimum value of ABDI for algal blooms exceeded the maximum values for clear water, turbid water, and thin cloud cover. In contrast, the values of FAI, AFAI, NDVI and EVI for algal blooms showed some overlaps with those for clear water, turbid water, or thin cloud cover. The ABDI value for thin cloud cover was low compared with those for the other indices, whereas the ABDI value for turbid water was lower than that of thin cloud cover and clear water. These results showed that the NDVI was most sensitive to turbid water and thin cloud cover, whereas the EVI and FAI were sensitive to turbid water. A previous study found that the AFAI performed well in detecting algal blooms using TM and MODIS data (Fang et al. 2018a). However, the present study found that the AFAI values for algal bloom showed some overlaps with those for turbid water and thin cloud cover. These results showed that the ABDI was less sensitive to turbid water and thin cloud cover compared with the other assessed spectral indices, and therefore that the ABDI could effectively extract information for algal blooms in Hulun Lake using MSI data.

3.3. Spatial resolution

The present study compared the areas of algal blooms extracted from MSI, OLI and MODIS data for Hulun Lake on 24 August 2019 to evaluate the importance of spatial resolution on the accuracy of the extracted algal bloom information (Figure 8). The ABDI was applied to MSI and MODIS-1,000 m, FAI was applied to OLI and MODIS-500 m and NDVI was applied to MODIS-250 m for the extraction of algal bloom information. Figure 8 shows that although information for algal blooms extracted

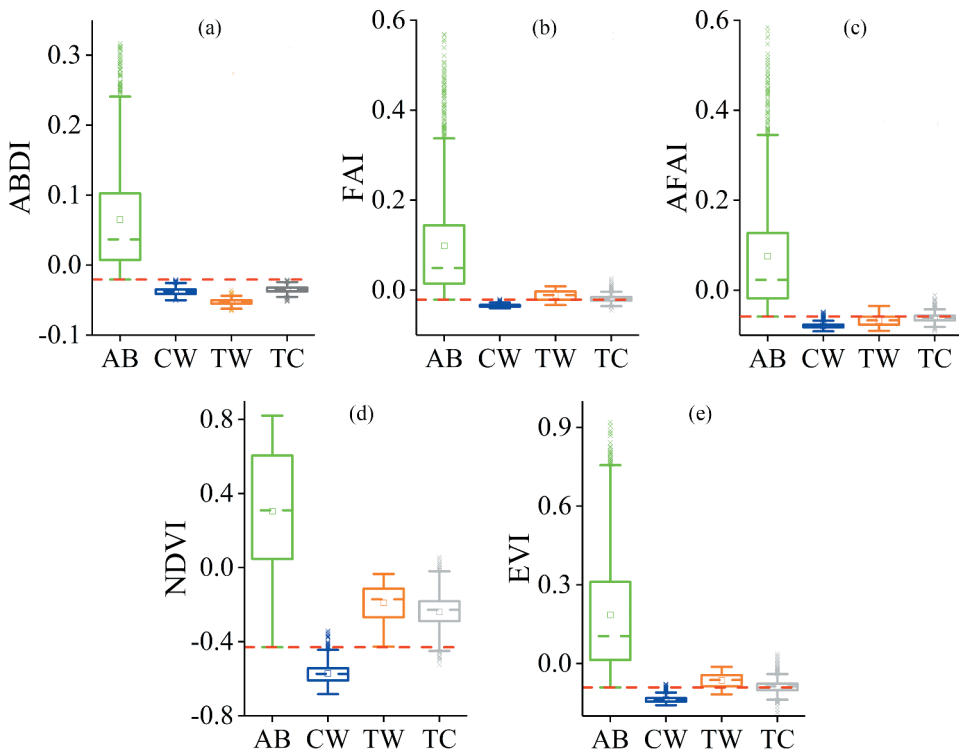


Figure 7. Box-and-whisker plots of values of ABDI (a), FAI (b), AFAI (c), NDVI (d) and EVI (e) for algal blooms (AB, $n = 2,582$), clear water (CW, $n = 1,323$), turbid water (TW, $n = 1,195$) and thin cloud cover (TC, $n = 1,044$) acquired on 24 and 31 August 2019 for Hulun Lake, China. Note that the red dashed line indicates the minimum value of algal blooms.

using the different sensors showed similar spatial distributions, the spatial distribution of algal blooms obtained through the MODIS image showed a low resolution, with details for smaller features unclear. In contrast, the MSI and OLI were able to capture the details for smaller features, such as small strips and patches of algal blooms. The effect of image resolution on uncertainty in the extraction of algal bloom information was further assessed by comparing the areas of algal blooms extracted using MSI, OLI and MODIS data with the results of visual interpretation. Table 3 shows the slope and R^2 of the relationships between the results of visual interpretation and the results of extraction from images of different resolutions. The results showed that MSI had the lowest uncertainty in the extracted information for algal blooms, obtaining the highest R^2 of 0.99 and slope of 0.93. OLI obtained a slightly larger uncertainty compared to MSI, obtaining an R^2 of 0.92 and slope of 0.84, which may be attributed to a coarser spatial resolution and the fact that information for algal blooms was extracted using the FAI. The use of MODIS data resulted in an overestimation of the area of algal blooms when compared to the results of visual interpretation maps. Uncertainties increased with decreasing spatial resolution from 250 m to 500 m to 1,000 m, with slope of 1.17, 1.84 and 2.22 and R^2 of 0.83, 0.81 and 0.78, respectively. This result can be attributed to MODIS data

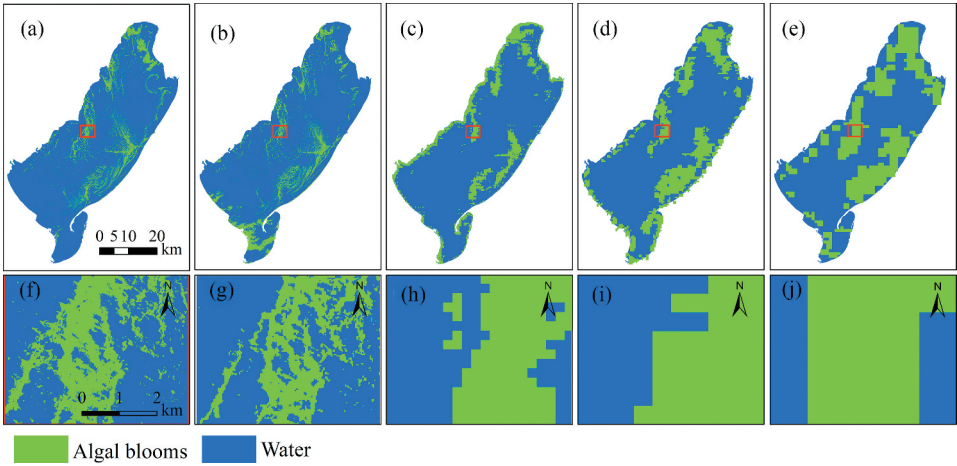


Figure 8. Comparison of results of areas of algal blooms extracted from images of different spatial resolutions taken on 24 August 2019 for Hulun Lake, China. (a) MSI with a spatial resolution of 10 m; (b) OLI with a spatial resolution of 30 m; (c) MODIS with a spatial resolution of 250 m; (d) MODIS with a spatial resolution of 500 m; (e) MODIS with a spatial resolution of 1,000 m. An enlarged map of the corresponding area (f–j). Note that red square is the location of the enlarged map.

Table 3. Slope and R^2 of the comparison between areas of algal blooms extracted from visual interpretation and that from spectral indices using sensors with different spatial resolutions. The images were acquired on 24 August 2019 for Hulun Lake, China.

Sensor	Spatial resolution (m)	Acquisition time (hh:mm:ss, GMT)	Slope	R^2
Sentinel-2 MSI	10	03:15:49	0.93	0.99
Landsat-8 OLI	30	02:57:29	0.84	0.92
MODIS	250	02:25:00	1.17	0.83
MODIS	500	02:25:00	1.84	0.81
MODIS	1,000	02:25:00	2.22	0.78

having relatively larger mixed pixel effects. These results indicated that the MSI data are suitable for the monitoring of algal blooms over small spatial scales.

4. Discussion

4.1. Atmospheric correction

Atmospheric correction constitutes a key step within the remote sensing of water colour, and directly affects the extraction of algal blooms. The present study conducted atmospheric correction of MSI images using the Sen2cor method. Atmospherically corrected MSI surface reflectance (MSI-SR) was evaluated using OLI surface reflectance products (OLI-SR), which have been successfully used as reference data (Vanhellemont and Ruddick 2014). Figure 9 shows a comparison of MSI-SR and OLI-SR data for 31 August 2019, whereas Table 4 shows the R^2 and RMSE of the relationship between the two datasets. Overall, the average MSI and OLI reflectance curves were similar in shape and magnitude (Figure 9 (a)), and the MSI-SR was in accordance with OLI-SR on 31 August 2019, with an R^2

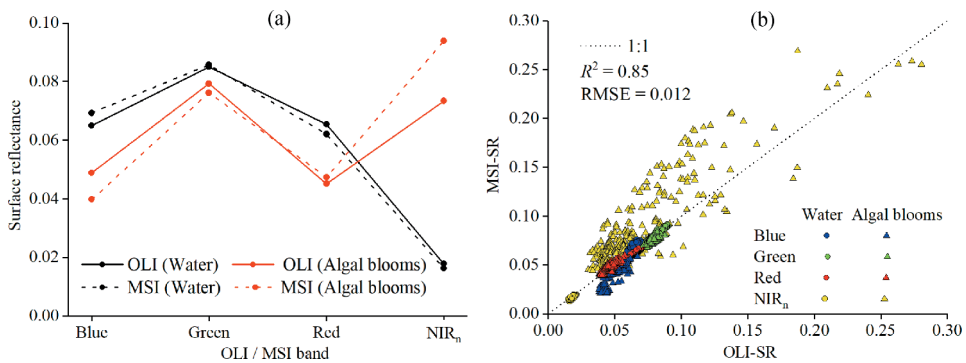


Figure 9. Comparison between the mean of MSI-SR spectra (dashed line) and that of OLI-SR spectra (solid line) on 31 August 2019 for Hulun Lake, China (a). Scatterplots between MSI-SR (y-axis) and OLI-SR (x-axis) on 31 August 2019 (b).

Table 4. R^2 and RMSE between MSI-SR and OLI-SR for Hulun Lake, China, acquired on 31 August 2019.

Date	Sensor and acquisition time (hh:mm:ss, GMT)	Sample type and number	Band	R^2	RMSE
31 August 2019	OLI (03:03:41) MSI (03:05:49)	Water ($n = 273$)	Blue	0.63	0.005
			Green	0.88	0.001
			Red	0.86	0.004
			NIR _n	0.77	0.002
		Algal blooms ($n = 250$)	Blue	0.64	0.011
			Green	0.33	0.004
			Red	0.75	0.003
			NIR _n	0.77	0.030

of 0.85 and RMSE of 0.012 (Figure 9 (b)). The consistency of reflectance for algal blooms between OLI and MSI was slightly lower than that for water, and some discrepancies between the reflectance of the blue bands ($R^2 = 0.63$ and 0.64) and green band ($R^2 = 0.33$) for algal blooms remained. These discrepancies may be mainly attributed to thin cloud cover over the lake waters (see Figure 5 (a)) and a difference in transit time between the two sensors. Fortunately, the reflectance of the blue band is not required within the calculation of ABDI, and the uncertainties in the green band, considered in part 2 of the ABDI, had little effect on the detection of algal blooms. Thus, these results show that MSI reflectance derived by the Sen2cor method can be applied for the detection of algal blooms in Hulun Lake.

4.2. The impact of meteorological factors on spatial and temporal changes in algal blooms in Hulun Lake

The ABDI was further applied to MSI images for analysis of spatial and temporal distributions of algal blooms in Hulun Lake. As shown in Figure 10, algal blooms in Hulun Lake appeared in summer and autumn (July, August and September) in 2019. Algal blooms developed in Hulun Lake in August, during which time the area of algal blooms increase rapidly of a short period of a few days. The majority of algal blooms were distributed in the littoral areas, particularly to the northeast and southwest of the lake. Figure 10 (b) shows a comparison between areas of algal bloom and temperature, precipitation, duration of sunshine and wind speed in 2019. Data from OLI and MODIS were concurrently analysed with MSI images within the extraction of area of algal blooms

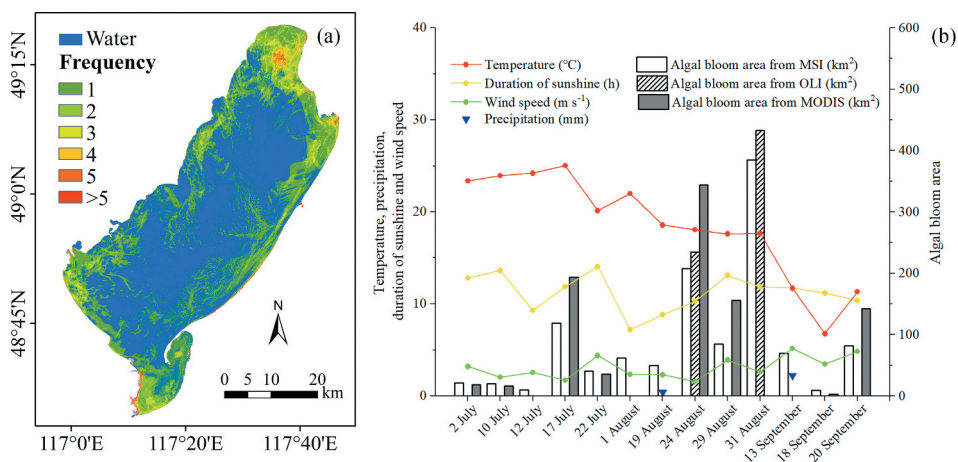


Figure 10. Frequency of algal blooms in Hulun Lake, China in 2019 (a) and the relationship between temporal variations in areas of algal blooms and meteorological factors in 2019 (b).

to validate the accuracy of area of algal bloom derived from MSI data. The results showed that areas of algal bloom derived through the MSI, OLI and MODIS had similar variations.

To date, no conclusive hypothesis exists to explain the relationship between meteorological factors and spatial and temporal changes in algal blooms due to the complex mechanisms of climate pressures on algal bloom. The present study conducted a preliminary analysis of the relationships between areas of algal blooms in Hulun Lake and meteorological factors, including temperature, precipitation, duration of sunshine and wind speed. As shown in Figure 10 (b), the average temperature on days with algal blooms in 2019 ranged between 6.8°C and 25°C, which indicated that algal blooms can adapt to a large temperature range. Although the average temperatures over the 19, 24, 29 and 31 August were similar (17.6°C to 18.6°C), a large difference in algal bloom area was evident (50.1 to 384.6 km²). The above situation also appeared on the 2, 10, 12 and 17 of July. Therefore, it is evident that temperature is not the key factor driving outbreaks of algal blooms in Hulun Lake and can rather be regarded as one of the fundamental conditions for an algal bloom outbreak (Fang et al. 2018c; Li et al. 2016). Some studies have suggested that precipitation inhibits occurrence of algal blooms, whereas others have concluded that algal blooms are promoted by precipitation through the transport of nutrients into the water body. However, since minor amounts of precipitation occurred on the 19 August (0.4 mm) and 13 September (2.2 mm), the present study could not conduct an effective analysis of this relationship, although it could be noted that areas of algal blooms on these 2 days were relatively small. The average duration of sunshine was 11.2 hours (h), with the highest of 14.1 h on the 22 July and the lowest of 7.2 h on the 1 August. Sunlight promotes photosynthesis within algal blooms and is therefore a necessary condition for their growth. The average wind speed on days with algal blooms was about 3.1 m s⁻¹, with a maximum and minimum of 5.2 m s⁻¹ and 1.7 m s⁻¹, respectively. For example, the average wind speed on the 17 July 2024 August and 31 August was only 2 m s⁻¹. Low wind speed has been shown to promote the converging and clumping together of algal blooms on the water surface (Xie et al. 2010). The above

results indicated that variations in algal blooms in Hulun Lake are driven by various meteorological factors. Human activities have also had great impacts on inshore waters, resulting in the accumulation of pollutants in lake waters. For example, industries fisheries (Yang et al. 2019), catering (Ding et al. 2011), agriculture and mining (Guo et al. 2019b) result in pollution of water bodies, which consequently accelerates eutrophication and algal blooms (Liang et al. 2016).

4.2. Applicability and limitations of the ABDI

The present study selected four regions in which algal blooms events are frequently detected to assess the applicability of the ABDI, namely the Chaohu, Taihu, and Dianchi lakes and the Yellow Sea. Cyanobacteria is known to dominate algal blooms in the Chaohu Taihu and Dianchi lakes, whereas green macroalgal blooms frequently occur in the Yellow Sea. Figure 11 shows the standard false colour composite MSI maps, the results of extraction of algal bloom area from visual interpretation, and those from the application of the ABDI and their scatterplots in the Chaohu, Taihu and Dianchi lakes and in the Yellow Sea. Table 5 shows the overall accuracies, K and commission/omission errors from the comparison. From the results, it was evident that the spatial distributions of algal blooms for the four regions extracted using ABDI were consistent with that extracted through visual interpretation, except for the eastern coastal areas of Taihu Lake which were covered by submerged aquatic vegetation (Ma, Tang, and Dai 2007). A comparison of the areas of algal blooms extracted through the ABDI and visual interpretation for the Chaohu, Dianchi and Taihu lakes and for the Yellow Sea showed an excellent linear relationship, with the scatterplots achieving a line close to 1:1, an R^2 greater than 0.97, and an RMSE less than 0.5 km², thereby demonstrating that the ABDI was effective in identifying algal blooms under a range of different conditions in different study areas. Table 5 shows that the overall accuracies of the ABDI for extracting the areas of algal blooms for the four regions were greater than 96%. However, the results for Taihu Lake showed a lower K of 0.70 compared to that of the other regions, and also showed the largest commission and omission errors of 27.76% and 29.74%, respectively. This result can be attributed to the possible presence of aquatic vegetation in the eastern part of Taihu Lake, which may have been misclassified as algal blooms due to the similarity in reflectance spectra with cyanobacteria blooms (Ma et al. 2008). These results implied that it may be difficult to distinguish between algal blooms and aquatic vegetation through the use of the ABDI in complex aquatic ecosystem in which aquatic vegetation and algal blooms coexist. In this situation, the accuracy of algal bloom extraction may be improved by combining the ABDI with other spectral indices.

5. Conclusions

The present study established an algal bloom detection index (ABDI) for the Sentinel-2 MSI sensor for the detection of algal blooms in Hulun Lake, China. Areas of algal blooms detected by the ABDI were in good agreement with that through visual interpretation, with the comparison achieving an R^2 of 0.87, RMSE of 0.67 km², an overall accuracy exceeding 98%, and a K exceeding 0.88. The ABDI showed lower sensitivity to thin cloud cover and turbid water compared to other spectral indices assessed, namely the

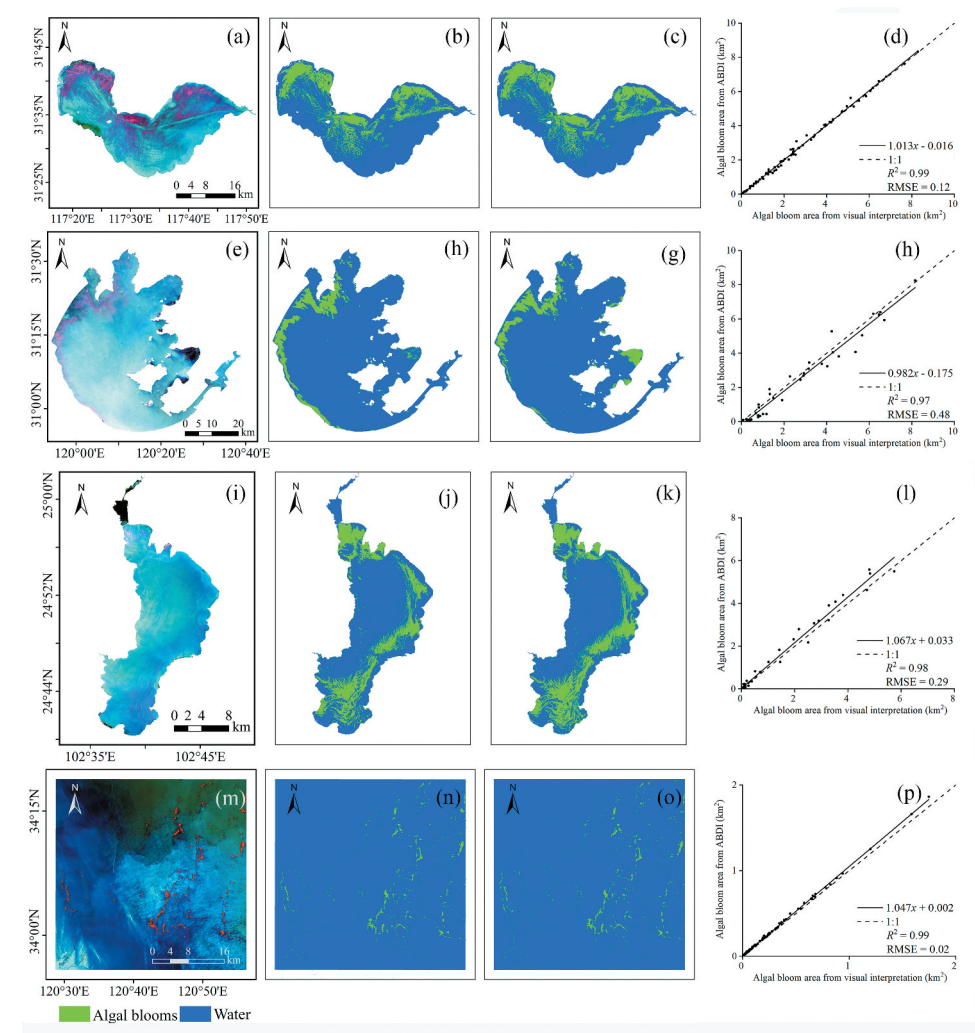


Figure 11. The general applicability of ABDI in different areas. Standard false colour composite (RGB: B8A/B4/B3) of Sentinel-2 MSI images in Chaohu (a), Taihu (e) and Dainchi lakes (i) and the Yellow Sea (m). Visual interpretation map in Chaohu (b), Taihu (f) and Dainchi lakes (j) and the Yellow Sea (n). The results of algal bloom extraction using ABDI in Chaohu (c), Taihu (g) and Dainchi lakes (k) and the Yellow Sea (o). Scatterplots between visual interpretation and ABDI in Chaohu (d), Taihu (h) and Dainchi lakes (l) and the Yellow Sea (p).

Table 5. The applicability of ABDI for the extraction of algal bloom area in the Chaohu, Taihu and Dainchi lakes and in the Yellow Sea.

Area	Overall accuracy, OA (%)	K	Commission error, CE (%)	Omission error, OE (%)
Chaohu Lake	99.52	0.97	2.54	2.07
Taihu Lake	97.80	0.70	27.76	29.74
Dianchi Lake	96.97	0.83	20.86	7.83
Yellow Sea	99.89	0.96	5.02	2.69

FAI, AFAI, NDVI and EVI. The ABDI was then applied to a time series of MSI images for 2019 to analyse the relationships between meteorological factors and spatio-temporal changes in areas of algal blooms in Hulun Lake. The results showed that outbreaks of algal blooms occurred mainly in August, with the drivers of algal blooms being identified as mainly temperature, precipitation, sunshine duration, wind speed, and human activities. In addition, it was found that the ABDI performed well in identifying areas of algal blooms under different environmental conditions through application to other regions, namely the Taihu, Chaohu and Dianchi lakes and the Yellow Sea.

Future research could investigate data measured in the field or data from images of high spatial resolution to further validate the accuracy of ABDI for identifying algal blooms. In addition, the ABDI could be applied to other sensors for the detection of algal blooms. Finally, further research on the influences of environmental factors on the dynamics of algal blooms in Hulun Lake could be conducted through obtaining further data on water level, air pressure, nitrogen and phosphorus content.

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Disclosure statement

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